

Section: Our Dynamic Universe

Topic: Collisions, Explosions and Impulse

A 2.0 kg object is moving at 6.0 m s^{-1} when it **collides head-on** with a stationary 3.0 kg object. After the collision, the two objects move off **together**.

(a) Calculate the velocity of the combined mass after the collision.

This is a **perfectly inelastic** collision (they stick together), so apply **conservation of momentum**:

$$m_1u_1 + m_2u_2 = (m_1 + m_2)v$$

$$(2.0)(6.0) + (3.0)(0) = (2.0 + 3.0)v \Rightarrow 12.0 = 5.0v \Rightarrow v = 2.4 \text{ m s}^{-1}$$

(b) Calculate the change in kinetic energy due to the collision.

Initial KE:

$$E_{\text{initial}} = \frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2 = \frac{1}{2} \times 2.0 \times 6.0^2 + 0 = 36.0 \text{ J}$$

Final KE:

$$E_{\text{final}} = \frac{1}{2} \times (2.0 + 3.0) \times 2.4^2 = \frac{1}{2} \times 5.0 \times 5.76 = 14.4 \text{ J}$$

Change in KE:

$$\Delta E = 36.0 - 14.4 = 21.6 \text{ J}$$

 Revision Tips

- If objects **stick together**, it's an **inelastic collision**
- Always apply conservation of **momentum**, not energy, in collisions
- Kinetic energy is only conserved in **elastic** collisions
- For total KE:

$$E_k = \frac{1}{2}mv^2$$

- Expect a **loss** in kinetic energy in inelastic collisions — often transformed into heat or deformation